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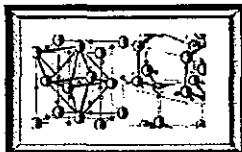
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COLD FUSION TIMES

ABC-TV REPORTS ON CF DEVICE

Gayle Verner

NEW YORK -- Readers will remember that in 1989, scientists Stanley Pons and Martin Fleischmann announced to the world that they discovered this new, self-sustaining, miracle dubbed "cold fusion". (Fusing hydrogen, which occurs on the sun, could be achieved with low temperatures rather than the million degree temps traditionally contemplated for "hot fusion"). But relentless criticism from many scientists may have incorrectly impugned the pair because the scientific process at that time was too difficult to reproduce. A number of other scientists persevered and were able to reproduce the excess heat. Slowly the pendulum reversed and the public saw occasional newspaper stories, pro and con, about the CF phenomena.

Now seven years later, in February '96, *ABC-TV's Good Morning America* featured a segment on the "Patterson device" that is said to get excess heat up to kilowatt levels. (continued page 3; discussions on pages 4, 6, 8)

POLITICAL INTEREST IN HELIUM HEATS UP

Cold fusion enables the harvesting of hydrogen contained in ordinary water to produce phenomenal quantities of energy. The most important aspect of cold fusion, other than direct energy production from matter, is the negligible pollution. What occurs is the generation of completely safe, and desirable, helium (He^4). The absence of pollutants which can alter the Earth's greenhouse effect or produce or alter the mainstream environmental setpoints makes this revolutionary energy - production process important. Helium is much more than just a lighter-than-air gas filler for party balloons.

As recently reported by Corey S. Powell in *Scientific American*, helium is used for a wide array of scientific and engineering purposes, including cryogenics and superconductivity, and for unexpected uses as even arc welding. Powell, along with the *American Physical Society*, has considered both strategic and political aspects of this matter.

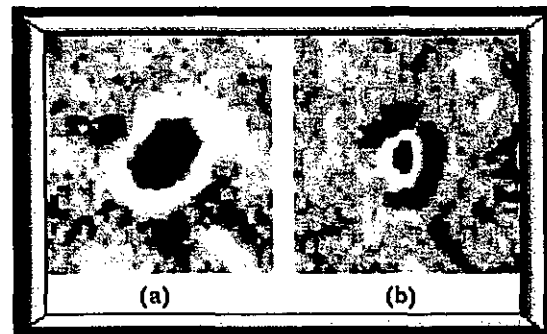
"It is also one of the earth's most limited resources, found in usable concentrations in just a handful of natural gas wells in the U.S. and Canada. Congress and the Clinton administration are acting to squander, rather than conserve, the helium supply, leading the *American Physical Society* to issue a warning against their "economically and technologically shortsighted" policies. Helium forms underground from alpha particles essentially the nuclei of helium atoms-emitted by radioactive elements in the earth's interior. (cont. page 5)

BOSE-EINSTEIN CONDENSATES AND COLD FUSION

Mitchell Swartz

Lasers provide coherent light. Bose-Einstein condensates (BEC) provide coherent matter -- a "matter laser". Conventional materials are made up of fermions which repel each other, by the Pauli exclusion principle which forbids simultaneously similar quantum numbers. Bosons -- in contrast -- have an attraction and similar energies and properties. They become a "superatom" and behave as if the entire cluster are in a single quantum mechanical state. Examples of bosons include photons and phonons. These remarkable condensates, named "Molecule of the Year -1995" (back cover), have figured in several theories involving cold fusion and other unusual material science.

Bosons have another unusual property. In 1925, Albert Einstein extended the work of Satendra Nath Bose and proposed that a theoretical gas of bosons would "condense" into a zero-kinetic state under certain conditions. (continued page 7)



Images showing a Bose-Einstein condensate
Clusters of lithium atoms (Li^7) before (a) and after (b) condensation. The spatial scale is 250 microns along each axis.
[a: 120,000 atoms at 590 nK (590 billionths of a degree above absolute zero), b: 20,000 atoms at 100 nK]—> More on pages 2 and 5.

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TOWARD THE FUTURE

[Ed. This excerpt concludes comments from Dr. Storms. The **COLD FUSION TIMES** continues to support Dr. Storms' solid reporting of scientific breakthroughs, which supplement the material and information previously presented in the **COLD FUSION TIMES**.]

CURRENT STATE OF COLD FUSION -- 1996

Edmund Storms

☑ The EUROPEAN UNION, after much study, has decided to issue the P-F patent. We all hope this will provide the push needed by the U.S. patent office to do what is increasingly obvious. If not, they will be taken to court for failing to follow case law. The situation has left the state-of-reasonable caution and has entered the state-of-scandal.

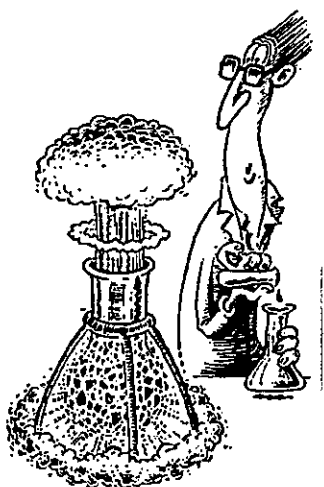
☑ The energy producing method discovered by Piantelli has been patented in Italy. He described his work at the ASTI meeting. A summary follows: The described cell contains a nickel rod surrounded along its length by a freestanding, single-layer coil of platinum wire. This is contained in a Pyrex tube into which hydrogen is introduced. The Pyrex assembly is contained in a metal envelope which is air-cooled. Thermocouples are located on the Ni rod, outside of the glass tube, and outside of the metal envelope. A current is passed through the Pt coil to heat the Ni, to calibrate the measured temperature with respect to power, and to provide a magnetic field. He claims this field is 2400 gauss but the value seems high. The system is heated to 700K, hydrogen is introduced at 0.2 atm, and the system is excited by various methods. These methods include changes in pressure or temperature. The magnitude of excess power is determined by noting the change in temperature measured at each thermocouple. The temperature change is in the range of 8-10 degrees. Claims were made for several observations. One claimed 29 W applied power resulting in 68 W excess for 218 days (1280 MJ) and the other claimed 20 W excess giving 590 MJ.

Use of deuterium instead of hydrogen produces similar powers while the system is said to turn on more quickly. Neutron emission was detected using a ^3He counter and gold activation. A steady background is seen with bursts when the conditions are changed. He claims an average flux of 600 neutrons/sec over a 3 week period. Clearly, a significant neutron flux is produced. Gamma emissions at 2.7 MeV, as well as several other energies, are seen.

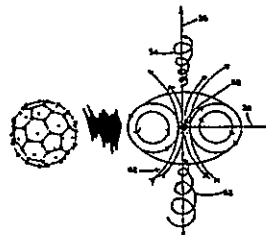
Clear evidence for charged particle emission having a range in air of @10 cm was seen when the Ni rod was placed in a cloud chamber. Large amounts of He^4 were detected but the measurements were not done well. The results have been reproduced in Italy using a nickel rod supplied by Piantelli. Unfortunately, he has not revealed the required pretreatment. His work is being funded by FIAT.

☑ Hideo Ikegami announced at the ASTI meeting that MITI has increased support for CF from \$30M/year to \$100M/year. Work in Japan has demonstrated the reality of CF and is now being directed toward a commercial product.

(continued page 3)



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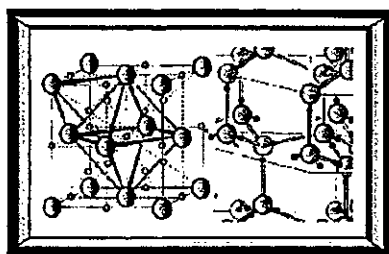
Cover Page: Bose-Einstein condensate after C.C. Bradley, et alia, description on page 5
Back Cover: Quantum portrait, Supercool Bose-condensed atoms peak near zero velocity. [M. R. Matthews]
Page 4: Mr. Patterson (*Infinite Energy Magazine*);

Cold Fusion Times LOGO -
Page 3 & 9: CETTS Beads (pedigree unknown) courtesy of John Logajan; SEM from reactions.
Palladium must be fully filled with deuterons to generate the reactions.
Scott Little

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CURRENT STATE OF COLD FUSION

(Continued from page 2)

Therefore, detailed information is no longer forthcoming.

☑ For the second time, FIAT arranged for a private workshop about CF in Italy. About 40 people from 4 countries were present. After a three-day discussion of the science, FIAT arranged for a news conference to announce their support for the field. At least this company is not shy about their support for the field.

☑ I have finally constructed a calorimeter and have used it to demonstrate excess energy. The sample used had the optimum properties based on a pretest of 19 samples. Detailed information about the nature of the

energy-active-state is being discovered and will be published soon. It's ironic that LANL will not support CF work even though it can now be done successfully in one's home.

☑ Work on CF is continuing in Russia in spite of their other problems. Excess energy using cavitation is getting a lot of attention and has resulted in a minor industry there.

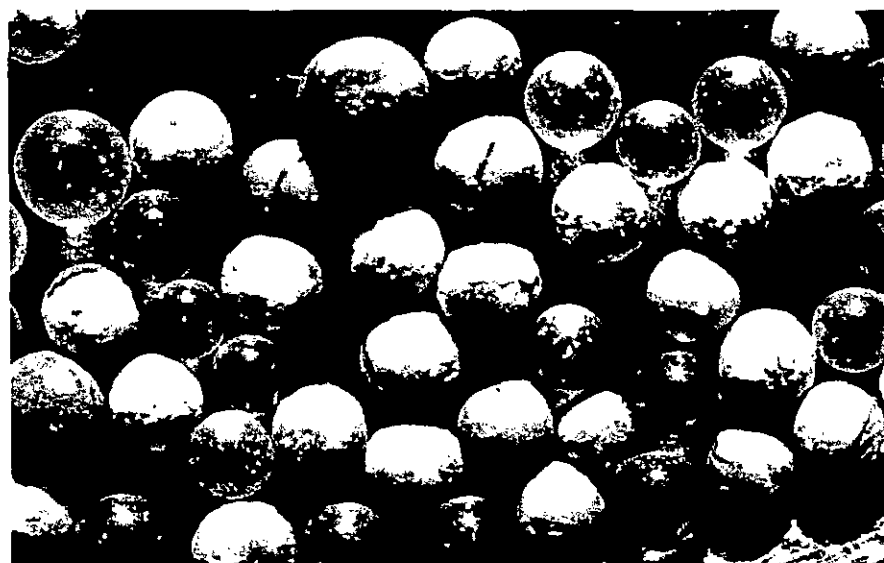
☑ The attitude of a skeptic is becoming more and more difficult to support without looking like a complete fool. Events are rapidly unfolding that will overwhelm such attitudes and will embarrass those who did not recognize the obvious in time. The field is gradually finding ways to go around the blocks to information exchange created by conventional scientific Journals and is finding support from commercial source. ➤

ABC-TV REPORTS

(continued from page 1)

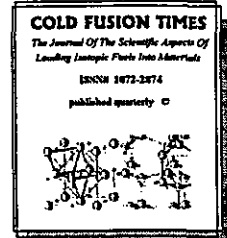
As reported in previous issues of the **COLD FUSION TIMES**, Prof. James Patterson, 71, invented the new cold fusion cell, has a US patent on it, and has made claims of its efficacy that apparently has attracted big time financial corporate giants such as *Motorola*. The idea, presented to the TV audience, came across as a nice neat package that ultimately will replace our oil burners, gas-powered cars and other fossil fuel driven items with low cost, safe, high energy, derived from water.

Although many TV viewers were "amazed" by the cold fusion potential, what the Patterson segment did not show was the proverbial "holes" potentially associated with it. Some scientists say that the measurements that Patterson has taken may be inflated, not properly gathered, or misinterpreted. As many know, a small controversy has erupted between true believers and skeptics of the Patterson device, and even among those that admit the cold fusion phenomena are real. While the publicity continues to generate, so does better scientific debate. ➤



Beads purportedly similar to *CETTs* showing heterogeneity, cracking, and degradation. Clear beads have been added in which are reportedly used for spacing. Pedigree unknown, origin anonymous, courtesy of John Logajan [Review of performance & function available **COLD FUSION TIMES** v4 n1]

OF GREAT INTEREST
TO OUR READERS
SEE FOR THE WORLD (PAGE 8)



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ABC's Coverage of the CETI Device

Transcript of ABC's GMAmerica program

[Ed. On February 7, 1996, ABC distinguished itself by probing into advances in cold fusion. **COLD FUSION TIMES** continues its coverage of the ABC discussion of the CETI demo here. The broadcast was reported by ABC Science editor Michael Guillen [Washington]; the transcripts are courtesy of Jed Rothwell and John Logajan]

Guillen: Thanks, Charlie. It is a device that its inventor says produces a hundred times more energy than it consumes. Now let me say right off the bat that lots of ideas come across my desk, that claim to be the energy source of the future, but this one is different. For one thing, the inventor has a distinguished track record. Second, the invention itself has been issued a patent by the US Patent Office. Furthermore, and this is key, independent scientists now claim to have reproduced the results, and major corporations like *Motorola* are taking a serious interest in it. So, is this potentially the greatest discovery since electricity? Since fire? Good question! Have a look.

James Reding, President, Clean Energy Technologies Inc: We have been able to reliably demonstrate a device that produces a thousand times more energy out than you put into it.



**Prof. George Miley,
University of Illinois:**

What could it do as far as an electric power plant or a water heater in your home? There are so many applications that the mind can run wild.

Dr. C. Quinton Bowles, University of Missouri: It would be a true source of power for use by the general public.

Guillen: It is hard to believe, but here is what is causing all the commotion. It doesn't look like much - some wires, some salt water, and at the core of it: this container of tiny beads. But these are no ordinary beads, and the man who invented them is no ordinary person.

James Patterson, CETI: I started making beads back in 1953.

Guillen: 74-year-old James Patterson looks about as homespun as his device, working out of a large garage in Sarasota, Florida, with more than a hundred patents to his credit. Patterson had always planned on being a chemistry professor but in 1951, while working for his PhD at Berkeley, *Dow Chemical* made him an offer he couldn't refuse.

Patterson: *Dow* hired me before I graduated, got my degree. And they paid me more than what I was going to get after I got my degree. So...

Guillen: It was during his years with *Dow* that Patterson invented a recipe for making tiny beads, beads so perfectly round that few people in the world can duplicate them.

Patterson: If I have a claim to fame, [laughs] I'm a good cook for little beads. Well, this is my storage area, and - it's almost like a library of what I've done.

Guillen: Over the years, Patterson's beads have been used in many different ways: in water purifiers, cosmetics, even as the 'talcum powder' inside surgical gloves.

Patterson: I'm better than a millionaire.

Guillen: Just because of the money you got from...

Patterson: ... little beads! [laughs] I have converted alchemy ... little beads into gold!

Guillen: Talk about alchemy! In creating his new energy device, Patterson took his regular beads, and coated them with thin layers of copper, nickel and palladium; a metal sandwich Patterson claims works like magic.

Patterson: This is the guts of it, this is creating heat.

Guillen: So this is water that you have flowing through it. [Feels outlet tube.] Oh, it is hot. Pretty warm! Yeah. And so how much energy is this little cell putting out, compared to what it's consuming?



James Patterson, CETI:

One watt. It's consuming only one watt, and it's putting out two hundred watts.

Guillen: You know this sounds too good to be true?

Patterson: [Laughs] Well, it may sound too good to be true, but if you'll only look, the scientific evidence is here. I mean, you're looking at it. I mean, you can't disavow what you're looking at.

Charlie: Michael, alright, we're looking at it, but how's it working?

Guillen: Well, you know, even for a scientist like myself, you can't just tell by looking or even touching it. That's where the other scientists come in, at the University of Missouri, the University of Illinois, and at *Motorola*. They have tested dozens of these devices, they say they can't get it not to work. Every time you plug it in, the doggone thing just produces all this excess heat.

Charlie: But what's going on, scientifically?

Guillen: Well, that's the big mystery. It's either, you know, an ordinary chemical reaction that's not behaving the way we expect it to, or some kind of a nuclear reaction. But there is no radioactivity that's evident from this thing so it doesn't appear to be a nuclear reaction. It's neither one nor the other, so it really is just a genuine mystery right now.

Charlie: Michael, what you are telling me is you have a scientific experiment that is producing a certain result and you have no idea how it's producing it.

Guillen: Yeah, but that's not unusual. I mean, very often times you run across something in the laboratory and you go 'wow! Look what it's doing' long before you understand why it's doing that.

Charlie: This sounds like the cold fusion debate again.

Guillen: Yeah. Remember the University of Utah, the whole cold fusion thing? Superficially this looks like cold fusion, in the sense that you have electricity passing through an electrode that is emersed in salt water. But there are essential technical differences. First of all the beads make this cell absolutely unique. That wasn't like the original cold fusion device. The other thing is that the original cold fusion device used heavy water, this uses ordinary water. So, it remains to be seen whether this is just a variation of the old cold fusion experiment or whether this is genuinely a new phenomena.

(continued page 8)

COLD FUSION TIMES <i>The Journal Of The Scientific Aspects Of Loading Isotopic Fuels Into Materials</i> Cold Fusion Times' Web Home Page: http://world.std.com/~mica/cft.html	COLD FUSION TIMES <i>The Journal Of The Scientific Aspects Of Loading Isotopic Fuels Into Materials</i> ISSN# 1072-2574 published quarterly ©
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MATERIALS CORNER

This issue: **BOSE CONDENSATES**

"MOLECULE OF THE YEAR"

(cont. from page 10)

experimentalists are rushing to create and explore this new phase, while theorists calculate its properties. Physicists are already angling to apply the new knowledge, hoping to capitalize on the condensate's unique aspects to create a laser that shoots beams of atoms instead of light. Understanding the laws that govern matter in this cold, coherent state may help physicists understand the mysteries of superconductivity and perhaps even the early universe. Physicists have glimpsed Bose-Einstein condensation before, but never in a system where they could study all its properties. For example, pairs of electrons and electron holes, known as excitons, have been observed to form condensates in semiconductors, but these last only a few millionths of a second. In ultracold liquid helium, up to 10% of the atoms are thought to be Bose-condensed, and this is intimately linked to startling properties such as superfluidity, which allows this fluid to creep up the sides of a beaker. But in liquid helium, the condensate is modified by the classical forces between atoms, so the quantum-mechanical signature is muddled.

To create a pure condensate, physicists need supercold atoms, where no heat veils the quantum forces. Yet the atoms must be kept in a gaseous phase and prevented from collapsing into a solid or liquid. In July, by cooling atoms of rubidium to within a whisper of absolute zero, researchers at a lab run jointly by the *National Institute of Standards and Technology* and the *University of Colorado* managed this feat. In an ultracold cloud of gas, they saw their boldest dreams come true, as a textbook example of Bose-Einstein condensation took shape. When they graphed the distribution of the atoms' velocities, they saw a now-classic portrait, with a dramatic peak close to zero velocity that represents the atoms slowly emerging from the condensate. The Colorado physicists slowed atoms with laser beams and trapped them in magnetic fields. They cooled their sample further with a method called evaporative cooling, which works much the way a cup of tea cools, by allowing the hottest atoms to escape. For although the first condensate was only about the size of a living cell, physicists are making larger samples all the time. In the fall, researchers at the *Massachusetts Institute of Technology* created a condensate with 200 times as many atoms as in the first sample, by modifying the magnetic trap with lasers and using sodium atoms. In principle, nearly three quarters of the elements in the periodic table could enter Bose condensation, so this year's research opens the floodgates for studies on a variety of different elements and conditions.

Because Bose-Einstein condensation is related to superconductivity, this work may help solve the mystery of how certain materials manage to conduct current without resistance. Also, some astrophysical theories invoke an asymmetrical pattern of Bose condensation to explain the bumpy distribution of matter in the early universe, a distribution that eventually led to the formation of galaxies. Having an actual Bose condensate could help develop such speculations.

In the long run, this new form of matter may yield substantial benefit to industry and society as well. That's because the condensate is to normal atoms as a laser is to a light bulb. In laser light, all the photons are in precisely the same phase, a property that makes laser

beams bright, powerful, and tightly focused. Similarly, the matter waves of atoms in a Bose-Einstein condensate are all in phase, and physicists are already working to design an "atom laser." Such a coherent beam of atoms could eventually pave the way for a new generation of incredibly fine manipulations and measurements. An atom laser might move atoms around one by one, or "write" atoms into semiconductors, much as lasers etch silicon chips today, but with much finer resolution.

M. H. Anderson et al., "Observation of a Bose-Einstein Condensation in a Dilute Atomic Vapor," *Science* 269, 198 (1995).

G. Taubes, "Physicists Create New State of Matter," *Science* 269, 152 (1995).

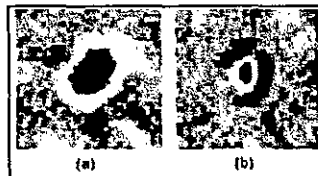
C. C. Bradley et al., "Evidence of Bose-Einstein Condensation in an Atomic Gas with Attractive Interactions," *Physical Review Letters* 75, 1687 (1995).

K. B. Davis et al., "Bose-Einstein Condensation in a Gas of Sodium Atoms," *Physical Review Letters* 75, 3969 (1995).

Cover Page: Images showing Bose-Einstein condensate

Images of trapped atom clouds resulting from different evaporative cooling cycles. These images were obtained by passing a probe laser beam through the atom cloud and recording the shadow of the atoms on a CCD camera.

Fig.(a) shows 120,000 atoms at 590 nanoKelvin (590 billionths of a degree above absolute zero), and (b) to 20,000 atoms at 100 nanoKelvin. The halo surrounding the absorption peak in (b) is due to diffraction of the probe laser beam from the spatially localized Bose-Einstein condensate. The condensate is roughly spherical in shape, with a diameter of about 3 microns.



From "Evidence of Bose-Einstein Condensation in an Atomic Gas with Attractive Interactions", C.C. Bradley, C.A. Sackett, J.J. Tollett, and R.G. Hulet, *Physical Review Letters* 75, 1687 (1995), and available at URL set to <http://atomcool.rice.edu/ccdpics.html>.

POLITICS OF HELIUM (cont. from page 1)

"Even more modest proposals—a natural-gas consumption fee that would finance a helium storage fund, for instance—face an uphill battle in the present political environment. Park is optimistic that a future Congress will at least restore the helium reserve. After all, maintaining it costs the U.S. only \$2 million a year, much of which is offset by money earned from short-term storage of helium for private providers. And when that reserve runs out? "We'll have to find new technologies," Park reflects. "Nature did not provide any substitute."

Obviously cold fusion offers much opportunity, even to replace this resource. ➤ - Mitchell Swartz

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THE PATENT PAGE

Analysis of the Intellectual Property

*** Returns Next Issue ***

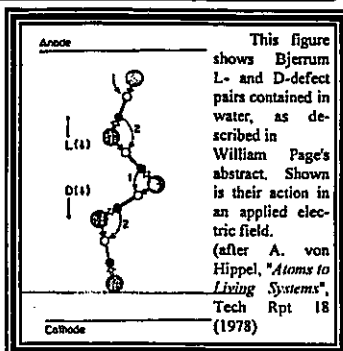
COLD FUSION THEORIES

Two-Dimensional Proton Conductors

William S. Page
Daneliuk & Page

In direct analogy with the treatment of electrons in conventional solid state physics, it is possible to describe the quantum motion of hydrogen in hydrogen-bonded solids and many liquids as a superposition of Bloch states. The detailed structure of the periodic potential that the hydrogen nuclei encounter is the result of the overlap of electron orbitals, but because the nuclei are so much more massive than electrons, for many purposes, we can represent the interaction between hydrogen and other atoms simply in terms of a generalized "Morse" potential. Just as for electrons, however, solutions of the wave equation take the form of energy bands. This has significant implications for the mechanical, thermal and electrical properties of hydrogen-bonded materials.

In particular, water (H_2O) in solid and liquid forms can be described as a proton semiconductor [Eigen (1958), Gosar & Pinter (1964), Fletcher (1970)]. Similar to the electrons in conventional semiconductors, in pure water almost all of the hydrogen is in the "valence" band state. Each oxygen atom has two loosely bound hydrogen atoms which form hybrid hydrogen "orbitals" with four tetrahedrally coordinated neighbouring oxygen atoms. Defects in this structure are classified as L-defects (empty hydrogen bonds, expressed in molecular terms as $OH^\cdot$) and D-defects (doubly occupied hydrogen bonds, expressed in molecular terms as H_3O^+). An L-defect can also be described as a "proton hole" in the valence band, while a D-defect corresponds to a proton in a conduction band. A gap of less than 1 eV separates the hydrogen valence band from the hydrogen conduction band. In comparison, the gap between the top of electron valence band and bottom of the electron conduction band is more than 10 eV. As a result, electrical conduction in pure water is almost entirely due to a small number of naturally occurring L/D-defect pairs created by thermally induced auto-dissociation. Water can be doped by donor or acceptor substitutional impurities. For



example, ammonia (NH_3) produces D-defects which is expressed in molecular terms as NH_4^+ and OH^- ions, while hydrogen fluoride produces L-type defects expressed as H_3O^+ and F^- ions. Water with an excess of D-defects is called a D-type electrolyte (analogous to N-type electron semiconductor). Water with an excess of L-defects is called an L-type electrolyte (analogous to P-type electron semiconductor). Both L-type and D-type electrolytes are good proton conductors. L/D-type junctions produce proton current diodes with properties similar to their electron counterparts. In particular such diodes display rectification (unidirectional current flow), high values of thermoelectric power, and may display luminescence under forward bias due to recombination of L and D-defects. Such effects as thermoluminescence of ice following irradiation and sono-luminescence of water may be related to L/D recombination. By the application of a potential difference at an interface, it is possible to produce an D-type inversion layer in an L-type electrolyte, again in direct analogy to an N-channel field effect transistor (N-MOSFET). Excess hydrogen is trapped in the one dimensional quantum well which lies parallel to the electrically negative interface (z direction), but free to move in the x and y directions. Well known quantum mechanical properties [Stern (1967)] apply to this inversion layer which are due to its two-dimensional nature and the periodic potential. The hydrogen atoms move with an "effective mass" which is approximately half that of a hydrogen atom outside the lattice. Further, as for a two dimensional electron gas, the density of energy states is a constant and the Fermi energy is proportional to the hydrogen atom density. Therefore, hydrogen density in the

NIGHTLINE ON "PATTERSON"

— A Review

by John Logajan

The *ABC-TV Nightline* program was a balanced look at the recent *CETI* claims with brief references to past cold fusion claims. Michael Guillen PhD, the Science Editor for *ABC-TV*, presented the documentary portion, which was an expanded version of that show earlier in the day on the *ABC-TV Good Morning America* show. Forrest Sawyer then conducted a debate between John Huizenga and Mike McKubre.

In the Guillen presentation, the skeptics were represented by Howard Birnbaum and Herman Feshbach. Patterson, Reding and their patent attorney Charles Prescott represented *CETI*. George Miley (*U of Ill*) speaking of his self-made Patterson Powercell said: "We consistently measured an excess energy coming out of it."

MG: "Does it always put out more energy than it takes in?"

George Miley: "Yes."

MG: "Does that surprise you?"

George Miley: "Absolutely. That's why we're so excited about trying to understand what's going on."

Quinton Bowles (*U of Missouri*, funded by Kansas City Power and Light) discussed the three cells they have operated over the last nine months, saying: "It's fair to say that all three of them appeared to be producing excess heat.... It works and we don't know why." Guillen said that Patterson had told him that *Motorola* had tested a cell and offered to buy Patterson out.

Sawyer's Huizenga/McKubre debate was a familiar rehash of standard form. One misstep by Huizenga was to admit to a Sawyer question that he hadn't reviewed the Patterson work carefully, but then a few sentences later Huizenga was asserting that these cells all use open calorimetry and fail to account for such things as recombination. Clearly since Huizenga had not studied the Patterson work, he was not in a position to categorize the calorimetry (especially since he was wrong about it.) McKubre attempted to correct Huizenga's false claim.

So in summary, the two major interesting bits, (continued page 8)

inversion layer increases with the strength of the applied potential and the kinetic energy of the hydrogen atoms increases more rapidly than Coulomb potential energy. The net result is to enhance the probability of inelastic scattering (including fusion) of the hydrogen nuclei. Such inversion layers may be present at the so called double-layer at the cathode in some electrochemical cells, beneath an oxide or silicate

layer on a cathode and on the surface of certain metals in hydrogen atmospheres. Two dimensional inversion layers can also be created in several types of layered crystal structures which display high proton conductivity such as, hydronium beta"-aluminate $H_2O-MgO.5Al_2O_3$ [Farrington (1978)], hydrogen uranyl phosphate $HUO_2PO_4 \cdot 4H_2O$, and sintered oxygen deficient oxides such as $SrCe_{0.9}Yb_{0.05}O_{3-\alpha}$. >

Eigen M., De Maeyer L. (1958) *Proc. Royal Soc. (London)*, Series A, 247, 505.
Farrington (1978) *Mat. Res. Bull.*, 13, 763.
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Stern F. (1967) *Phys. Rev.* 163 (3), 816.



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COLD FUSION TIMES

ENGINEERING AND MATERIALS VIEW TOWARD COLD FUSION.

THIS ISSUE: BOSE-EINSTEIN CONDENSATION

JET Technology

[Ed. The excerpted articles will bring our readers up to speed on this important subject. Below is a joint release by the *U. of Colorado*, and the *NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)*. *JILA* is the Joint Institute for Laboratory Astrophysics. Images of this new state of matter are shown on pages 1, 5 and 10.]

PHYSICISTS CREATE NEW STATE OF MATTER

Boulder, Colorado - Physicists have achieved a temperature far lower than has ever been produced before and created an entirely new state of matter predicted decades ago by Albert Einstein and Indian physicist Satyendra Nath Bose. Cooling rubidium atoms to less than 170 billionths of a degree above absolute zero caused the individual atoms to condense into a "superatom" behaving as a single entity, said Eric Cornell and Carl Wieman of *JILA*, a joint program of the Commerce Department's *National Institute of Standards and Technology* and the *University of Colorado* at Boulder. Before photographing the superatom, the physicists cooled the atoms to 20 billionths of a degree above absolute zero, the lowest temperature ever achieved.

Scientists have sought to create the effect for more than 15 years. The team used laser and magnetic traps to create the Bose-Einstein condensate, a tiny ball of rubidium atoms which are as stationary as the laws of quantum mechanics permit. This ball is surrounded by a diffuse cloud of normal rubidium atoms. Made visible by a video camera, the condensate looks like the pit in a cherry except that it measures only about 20 microns in diameter or about one-fifth the thickness of a sheet of paper. The condensate was formed inside a small glass cell surrounded by a tabletop array of magnets, lasers and computers in a *JILA* laboratory. Within the atom trap are about 2,000 atoms of rubidium squeezed into a very small space at a very low temperature. The atoms within the condensate obey the laws of quantum physics and are fundamentally different from the normal atoms in the much less dense cloud surrounding it. The *JILA* team cooled the atoms to a temperature 300 times lower than has ever been achieved in other scientific laboratories. "Atoms in a room temperature gas normally move about 1,000 miles per hour and slow down as temperatures drop," Cornell said. "The normal atoms at these low temperatures move about 3 feet per hour. The Bose-Einstein condensate atoms move a lot slower, too slow for us to measure yet."

Wieman's tactics in pursuing the condensation initially were met with skepticism in the scientific community. But as his and Cornell's methods began to show that the goal was achievable, several other teams of physicists joined the chase.

Beginning with a gas of room-temperature atoms, the *JILA* team first slowed the rubidium and captured it in a trap created by the light produced by diode lasers similar to ones used in compact disk players. The infrared lasers are aligned so that the atoms are bombarded by a steady stream of photons from all directions -- front, back, left, right, up and down. The wavelength of the photons is chosen so that they will interact only with atoms that are moving toward the photons. For the atoms, "It's like running in a hail storm so

BOSE-EINSTEIN CONDENSATES AND COLD FUSION

(Continued from page 1)

The requirement would be that the de Broglie wavelength of the bosons would exceed the average interparticle distance within the gas. What followed was that the Bose-Einstein condensate is believed to be responsible for superconductivity, and for the superfluidity of helium 4 [whereby this material will climb up the walls of a container in seeming defiance of the laws of gravity]. Bose condensation has even been invoked to have played a role in the Big Bang wherein the Higgs boson metamorphosed by BEC to the universe which surrounds, and has become, us.

However, the BEC was not actually found for decades. The elusive search for the Bose-Einstein condensate (BEC) was achieved for alkali metals only at temperatures as low as a few microkelvin. This spectacularly cold condition -- millions of times colder than interstellar space -- was achieved by illuminating clusters of alkali atoms from all directions simultaneously so as to chill the molecular clusters to a millionth of one degree above absolute zero.

**Bose condensation may be the origin
both of the cold fusion phenomena and
the unexpected reaction products**

Could Bose
condensation play a
role after loading?
Cold fusion, an engi-

neering feat undreamed of ten years ago, offers fantastic opportunities for the coming century. By a variety of techniques, including electrolysis and gas loading, special types of metals can become filled with the hydrogen to concentrations not achievable anywhere else, and if the loading exceeds ~85%, then the excess heat (enthalpy) results.

Such extended quantum mechanical particles as BEC may, in fact, be the etiology both for the cold fusion phenomena, and the tardive acceptance of these solid state/nuclear phenomena by mainstream physicists and some scientists. In a quantum mechanical sense, wavefunctions of the fuels loaded into the materials, may become condensed Bose states, and ultimately permit fusion within the solid. The Bose condensation states exhibit the propensity for change, which under some conditions, which may produce the desired reactions yet bypass the conventional reaction products. ➤

that no matter what direction you run the hail is always hitting you in the face," Wieman said. "So you stop." This cools the atoms to about 10 millionths of a degree above absolute zero, still far too hot to produce Bose-Einstein condensation, and about 10 million of these cold atoms are captured in the light trap. Once the atoms are trapped, the lasers are turned off and the atoms are kept in place by a magnetic field. All atoms have a tiny magnet attached to them caused by the spin of the electron. The atoms can be trapped, or held in place, if a magnetic field is properly arranged around them, the researchers said. The atoms are further cooled in the magnetic trap by selecting the hottest atoms and kicking them out of the trap. It works in a way similar to the evaporative cooling process that cools a hot cup of coffee -- the hottest atoms leap out of the cup as steam. ➤

NIGHTLINE -- REVIEWED

(continued from page 6)

in my opinion, were that Miley at the U of Ill and Bowles at the U of Missouri said in their own words that they had independently seen excess power in Patterson cells under their own control. Guillen said they are reviewing the results before they attempt to publish in the standard periodicals. >

ABC GOOD MORNING AMERICA TRANSCRIPT

(cont. from page 4)

Charlie: Is there an anticipation that what is taking place here in microcosm can take place in a macro situation where you can produce a tremendous amount of energy?

Guillen: Now *that* is going to be the key question. If the scientists at the independent universities and corporations continue to verify that this device seems to work, the next question is going to be: can you scale it up from this laboratory model into something that can be mass produced, and be cost efficient. Because we have heard other alternative energy like wind power and solar power, they also sound great but they have never become cost efficient. That's going to be the big question in the future.

Charlie: You keep saying 'if this works.' You are telling me that a number of scientists have been able to make it work. There are also a bunch of other scientists who are saying this is just crazy.

Guillen: Yeah. The scientists are really cautious because of the old cold fusion flap six years ago. They want to be real cautious. The question is here, you have to measure the temperature differences, how much of the heat is putting out . . . is being put out by this device. That requires you to use thermometers of various kinds. They are just double, triple and quadruple checking those thermometers to make sure they are not misreading them. But they are all saying yes, this seems to work as advertised. So it's potentially historic.

FOR THE WORLD

JET Technology

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(More Information -- See Pages 2 And 3 Too)

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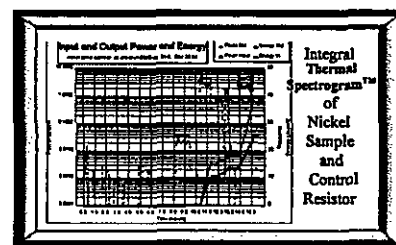
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MORE COMMENTS ON THE CETI MICROSPHERE DEMO

There were many intriguing responses to the CETI review last month, and we may cover some in coming months. Here, Prof. Dennis Cravens of CETI is given the opportunity to respond to his critic, Prof. Steven Jones, on the issue of Recombination as the origin of the excess heat for cold fusion in general, and for the CETI microsphere demo in particular.

Comments on Steve Jones' Theory on Recombination

Dr. Dennis Cravens CETI, Clean Energy Technology, Inc.

I will start by noting that Jones is usually a careful experimenter (in the nuclear measurement area) and makes some interesting points. However, it is obvious from the response that he neither attended the demo at the SOFE conference nor is in possession of some information to make a correct interpretation of the event. First, the demo was a demo and not a detailed experiment. The system was assembled at U of I by one of their grad students. Its primary function was to obtain qualitative measures of different bead compositions. The reason that Miley does not reveal the construction of the beads he is preparing is that those are part of an ongoing R and D by CETI. The beads used Mon., Tues. and Wed. were chemically prepared by CETI (Patterson). Those were fully disclosed and their detailed construction and preparation are in the four granted patents. At the SOFE conf. a diagram of the beads cross section was displayed and copies of the patents were available for anyone wishing to make the patented beads for research purposes. The beads used later Wed. were prepared by the U of I. This technology and some production techniques are not yet patented and were not fully disclosed. Though Jones tries to imply that not telling everything is somehow wrong, I see nothing wrong with not disclosing techniques and research until after the experiments have been conducted and analyzed. Unlike Jones I do not think that ongoing research nor grant proposals should be made public (referees should not make public reviews of pending grant proposals). The bottom line is that if you want to know about the beads on which most experiments have been conducted and my report was written on, then it is available in the patents and was described in my ICCF-5 presentation.

I will freely admit those areas where there may be problems. Jones mentions the problems with the temperature measurements but dismisses it. I feel and have freely admitted that if there is a real problem it is there. In the ICCF-5 paper, I list the problems and the corrections connected with the thermocouples in the systems. Yes, you can sometimes get into trouble with thermocouples in electrolytic cells. This is why my paper mentions that the thermocouples were moved to externally electrically insulated stainless steel wells and the wells were connected to a common ground. Further, thermistors were used to verify their reading and finally mercury thermometers were used as a "reality check". Jones has either not read the work on the system he is criticizing or is willfully trying to mislead others in believing such concerns have not been addressed. I personally think that if there is a real problem with the temperature measurement it is due to thermocouple placement. That is why calorimetry on current systems includes, in-line mixers, checks with thermistors, RTD's and glass thermometers.

A major problem with this whole field is that people scurry around trying to dig up information about the ongoing research without having access nor taking the time to wait for the full and finalized experiments. If you are going to criticize an experiment for calorimetry, then it should be on an experiment that was designed for such measurements and not a device designed to give qualitative variations in materials destined for those more detailed experiments. You should wait for the experiment and its write up on the experiment.

Jones' idea of excess coming from recombination is not tenable. First it should be mentioned that the demo at SOFE was run for 12 hours on Mon (8-8) and about 13 hrs on Tues. in full view of everyone. Some people even checked meters and substituted meters. The device ran with only about 200ml of electrolyte (total in the system including tubes, filter, etc. with only about 5ml in the cell itself). The current during

OP-ED COLUMN -- WHAT'S HAPPENIN' Spring '96 and NEWS from the INTERNET (also page 3 and 6)

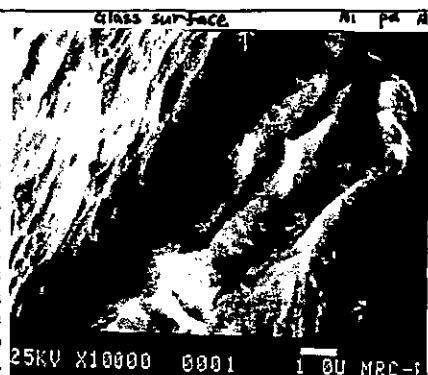
the entire running time was set at 0.02A and the voltage ranged from 2.9 to 3.2V (normally 3.0V). The device was turned off overnight and it took about 30 min. to get started each morning. During Mon. and Tues. the thermal output was between 3.5 and 6W and averaged around 4.5W. (the water flow was 14 to 20 ml/min.)

Jones' idea that the 30 min. at 0.06W of input (0.03 watt hours of energy) could somehow be responsible for the 12 hours at 4.5W (54 watt-hours) violates the conservation of energy. His idea that the hydrogen and oxygen are somehow stored in the electrolyte or cell and by recombination are responsible for the 4.5W is also invalid.

First notice that the electrical input to the device was 0.02A and normally 3V or 0.06W. Only half of that is available for the production of gas (0.02 times the 1.5V required to dissociate the water) and the other half goes to heating the solution. Thus if all of the hydrogen and oxygen recombined you could only get about 0.03W of heat from that process and 0.03W from joule heating of the electrolyte. You would still be missing out about 4.4W of thermal power. Second the idea that the electrolyte could store the hydrogen and oxygen to be later recombined to give the 4.5W is also not consistent with the reality of the demo. The 200ml of electrolyte would have been required to store 54 watt-hours or 27 liters (for Mon. alone and if you believe that no one pumped hydrogen and oxygen gases into it Mon. night during the discussion you would need a total of 110 W-hr or 55 liters) of gases (18L of hydrogen and 9L of oxygen) stored in the 200ml of the electrolyte. The pressure over the open cell was only 1atm. I have checked and can only dissolve about 7ml of oxygen in 200ml of the electrolyte at saturation (when cool). The concept of placing a frit to bubble argon through the cell is not physically possible as anyone seeing the demonstration would have realized. Even in cells with large electrolyte volumes, the general idea of bubbling argon through a cell to purge oxygen and hydrogen is not as simple as Jones suggests. The difficulty comes from the calorimetry. The cooling of the electrolyte due to expansion of the gas, controlling and measuring the gas flow, and the loss of heat due to the heat of vaporization of water (evaporation assisted by the argon flow) would be very hard to measure in practice. His "simple" method would drastically underestimate the heat production unless the cooling from the gas effects could be monitored. Thus the experiment as Jones described is not consistent with known calorimetric procedure. I can see no reason for expending the time and money for such experiment unless Jones can tell me how the 200ml of electrolyte could hold 27 to 55L of hydrogen and oxygen at 1 atm instead of the 7ml that I measure. >

Scanning Electron Microphotograph of Trilaminar Bead

Scanning Electron Microscopic image of Ni/Pd/Ni coated glass beads from Scott Little (EarthTech, Intl.). The magnification is 10,000x. The three layers are ~1 micron thick. Apparent separation of layers from glass substrate (large area on the left) may be from the grinding process required to section the bead for microscopy.



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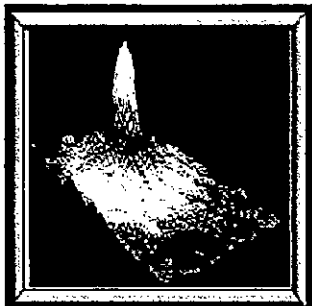
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[Ed. Readers of the **Cold Fusion Times** have been reading about Bose-Einstein condensates for several years. This new state of matter is critical to this, and other, technologies, and has been named "Molecule of the Year" by the *AAAS (American Academy for the Advancement of Science)*, as cited in this excerpt]

BOSE-EINSTEIN CONDENSATE

NAMED MOLECULE OF THE YEAR

Back in 1924, Albert Einstein predicted the existence of a new phase of matter, an exotic state in which atoms defied

the laws of classical physics and followed only the dictums of quantum mechanics. In 1995, by chilling wisps of gas to ultracold temperatures, physicists finally got their first good look at this state, named the Bose-Einstein condensate after Einstein and Indian physicist Satyendra Bose. This year's work ends an arduous quest and ushers in a new age of exploration in atomic and condensed-matter physics. We salute the condensate as Molecule of the Year for 1995, but this peculiar form of matter is not a molecule. Indeed, this year's magnificent achievement was to elude the everyday forces that bind atoms together into molecules and so unmask the more subtle powers of quantum mechanics. While atoms in an ordinary gas dart about in all directions, the atoms in the condensate move in lock step, at identical speed and direction. They have relinquished their individual identities to become a single, collective entity, and their organized condition is expected to give rise to bizarre properties. The condensate's unusual nature makes it an ideal workshop for exploring the counterintuitive realm of quantum mechanics. So 6 months after the first dramatic report, (continued on page 7, see also page 5)

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